

TrES-3b

R-2255

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_120519 · created 2026-08-16T00:34:29+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456066.8211 +/- 0.00085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.013
Transit depth (Rp/Rs)^2	3.36 +/- 0.49 [%]
Orbital Inclination (inc)	82.52 +/- 0.51
Airmass coefficient 1 (a1)	1.247 +/- 0.013
Airmass coefficient 2 (a2)	-0.197 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	None
Optimal Aperture	3.6
Optimal Annulus	5.11
Transit Duration (day)	0.0626 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.013 vs 0.16309 (deviation 1.53σ)
Duration fit vs published	0.0626 d vs 0.0567 d (deviation 1.41σ)
Mid-time O–C	1.8 min
Depth SNR	14.1
Residual scatter	1.01 %

Light curve

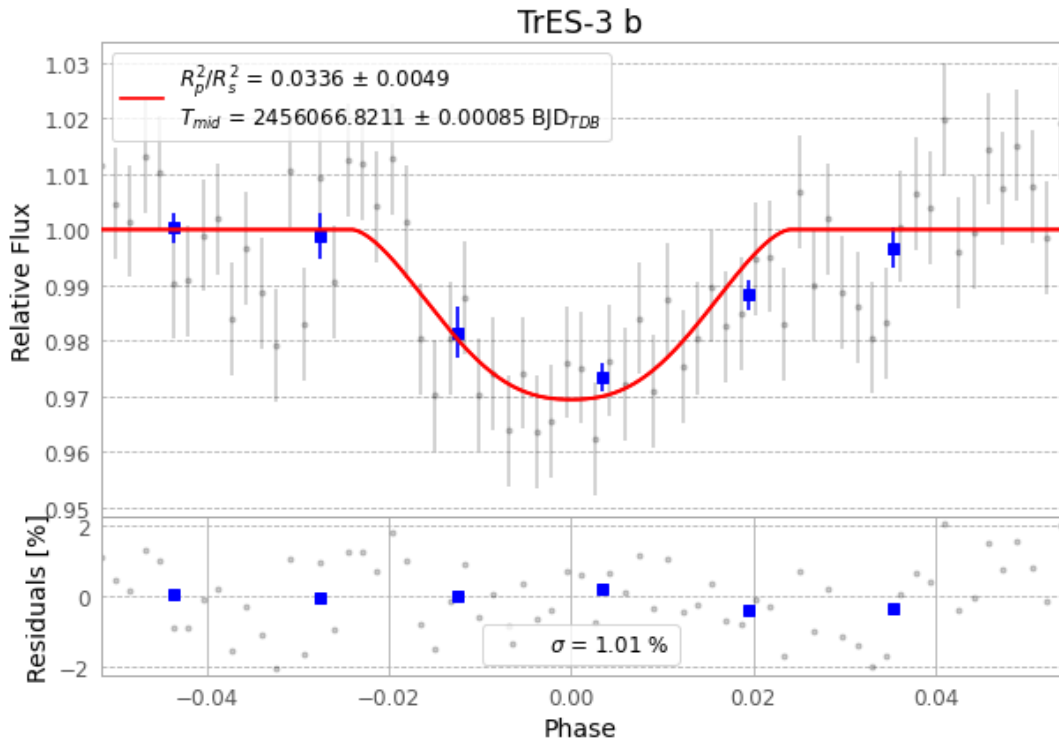


Figure 1 — FinalLightCurve_TrES-3 b_2012-05-18.png (SHA-256 f403053b0891f524...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 191], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 93bbd8548a212519...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

67 FITS frames (44 MB), TRES-3120519060012.FITS → TRES-3120519091814.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120519.log (6c41196e1eb8...), FinalLightCurve_TrES-3 b_2012-05-18.png (f403053b0891...), FinalLightCurve_TrES-3 b_2012-05-18.csv (c0a6e5cc01e6...)

Compute resources

Wall time 112.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 00:34Z.